Weight-space gradient descent induces approximate activity descent, making learning rules testable

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The problem: synaptic learning rules are hidden, but activity changes are not. A central question in neuroscience is whether biological neural networks optimise a loss function by adjusting synaptic weights—and if so, by what rule. Decades of theoretical work have proposed candidate learning rules. But synaptic weights cannot be measured at scale in behaving animals, so these proposals remain difficult to test directly. Neural *activities*, by contrast, can be recorded in large populations across learning. This raises a concrete question: if the brain does perform gradient descent on its weights, what should we expect to see in the activities? Here we show that the answer takes a surprisingly simple form—activity changes must be approximately proportional to activity-space loss gradients—and that this prediction is directly testable with existing experimental techniques.

Setup and derivation. Consider a feedforward network with weights $\mathbf{W}$ and neural activities $A_i(\mathbf{W})$, trained to minimise a loss L. Gradient descent on the weights embodies a simple principle: *the more that strengthening a synapse would improve performance, the more that synapse is strengthened* (and vice versa—harmful synapses are weakened). Formally, every weight is updated simultaneously:

$$\Delta W_\alpha = -\eta \, \frac{\partial L}{\partial W_\alpha} \,, \qquad \forall \, \alpha. \tag{1}$$

What is the induced change in the activity of neuron i? Because A_i is a deterministic function of all weights, we can apply the chain rule over the full parameter vector:

$$\Delta A_i \;=\; \sum_\alpha \frac{\partial A_i}{\partial W_\alpha} \, \Delta W_\alpha \;=\; -\eta \sum_\alpha \frac{\partial A_i}{\partial W_\alpha} \, \frac{\partial L}{\partial W_\alpha} \,. \tag{2}$$

Expanding $\partial L/\partial W_\alpha = \sum_k (\partial L/\partial A_k)\,(\partial A_k/\partial W_\alpha)$ and exchanging the sums yields

$$\boxed{\Delta A_i \;=\; -\eta \sum_k \Theta_{ik} \, \frac{\partial L}{\partial A_k} \,,} \tag{3}$$

where

$$\Theta_{ik} \;\coloneqq\; \sum_\alpha \frac{\partial A_i}{\partial W_\alpha} \, \frac{\partial A_k}{\partial W_\alpha} \;=\; \nabla_{\mathbf{W}} A_i \cdot \nabla_{\mathbf{W}} A_k \tag{4}$$

is a neural-tangent-kernel-style Gram matrix defined on internal activities rather than network outputs. In vector form, $\Delta \mathbf{A} = -\eta \, \Theta \, \nabla_{\mathbf{A}} L$: gradient descent in weight space induces *kernel descent* in activity space, with Θ encoding how strongly every pair of neurons shares upstream parameters. This identity is exact and holds for any differentiable network, regardless of depth, width, or activation function.

The self term drives each neuron toward its own loss gradient. The diagonal element $\Theta_{ii} = \|\nabla_{\mathbf{W}} A_i\|^2$ collects contributions from *all* weights that influence neuron i—both its own incoming weights and every weight in earlier layers that lies on a path to it. Because $\Theta_{ii} > 0$, the self term is always aligned with $-\partial L/\partial A_i$: each neuron's activity moves down its own loss gradient, with an effective learning rate $\eta \, \Theta_{ii}$ that grows with the depth and width of its input pathways. In other words, weight-space gradient descent guarantees that every neuron has a built-in tendency to change its activity in the loss-reducing direction.

Cross terms introduce interference from other neurons. The off-diagonal elements $\Theta_{ik} = \nabla_{\mathbf{W}} A_i \cdot \nabla_{\mathbf{W}} A_k$ for $k \neq i$ couple neuron i to every other neuron's loss gradient. Their magnitude reflects parameter sharing: two neurons fed by overlapping early-layer weights will have large $|\Theta_{ik}|$; neurons in disjoint pathways will have $\Theta_{ik} \approx 0$. These cross terms have no reason to be aligned with $-\partial L/\partial A_i$ and constitute interference—the side-effect of the rest of the network learning simultaneously. Whether the self term or the cross terms dominate determines whether activity changes look like activity-space gradient descent or something messier.

Wide networks make the kernel diagonal-dominant. The cross terms vanish when $\nabla_{\mathbf{W}} A_i \cdot \nabla_{\mathbf{W}} A_k \approx 0$ for $i \neq k$—that is, when different neurons' activities depend on nearly orthogonal directions in weight space. Several settings push the kernel toward this regime.

Width. At random initialization, two neurons in the same layer share no incoming weights, so their activity gradients are independent random vectors in parameter space. By concentration of measure, their inner product scales as $O(1/\sqrt{d})$ while the diagonal scales as $O(1)$, making Θ increasingly diagonal-dominant as width grows—the internal-activity analogue of the NTK becoming deterministic at infinite width.

Depth and weight sharing. Neurons in the same layer share all earlier-layer weights, which introduces a common component into their parameter gradients. Each neuron's own incoming weights act as independent random projections that decorrelate this shared component in the wide limit. Deeper networks have more shared parameters but also more such projections; in practice, the cross terms remain significant at finite width.

Learning dynamics. Gradient descent itself may reinforce diagonal dominance. Training is known to decorrelate neural representations as the network learns to tile the input space efficiently. To the extent that decorrelated activities imply decorrelated parameter gradients—since $\partial A_i/\partial W_\alpha$ depends on upstream activations feeding into i—learning should progressively shrink the off-diagonal entries of Θ. The link between activity and gradient decorrelation is heuristic and deserves formal treatment, but it suggests that trained networks may be closer to the diagonal regime than their initializations.

Taken together, these arguments imply that in sufficiently wide networks—and cortical circuits are very wide—the kernel Θ is approximately diagonal, and activity changes closely track the negative activity-space loss gradient.

The diagonal approximation: activity descent as a proxy for weight descent. Combining the kernel identity (Eq. 3) with diagonal dominance ($\Theta_{ii} \gg \sum_{k\neq i} |\Theta_{ik}|$) yields a simple approximation:

$$\Delta A_i \;\approx\; -\eta\,\Theta_{ii}\,\frac{\partial L}{\partial A_i}\,. \tag{5}$$

The intuitive meaning mirrors that of weight descent: just as weight descent says *strengthen each synapse in proportion to how much it helps*, activity descent says *increase each neuron's firing in proportion to how much that increase would help* (and decrease it when it hurts). Equation 5 shows that one implies the other in wide networks—the per-neuron scaling Θ_{ii} may differ across neurons, but every neuron's activity moves in the direction that locally reduces the loss. This is the paper's central result: *if a network performs gradient descent on its weights, and the network is wide enough, then its activity changes must approximately follow gradient descent in activity space.* The approximation improves with width and, as our simulations show (Figs. 1–2), is already accurate at moderate widths.

A testable prediction for neuroscience. This result converts an untestable claim about synaptic learning rules into a testable prediction about observable activity changes. Synaptic weights cannot be measured at scale in vivo, but neural activities can. If a biological circuit performs *any* form of gradient descent on its synaptic weights—even an approximate or noisy variant—and is sufficiently wide that Θ is diagonal-dominant, then the observable activity changes must satisfy Eq. 5. The per-neuron scaling Θ_{ii} is unknown and may vary across neurons, but the *direction* of $\Delta\mathbf{A}$ is pinned: it must be approximately proportional to $-\nabla_{\mathbf{A}} L$. A concrete experimental protocol follows: record neural population activity across learning trials, estimate $\partial L/\partial A_i$ from a task-fitted model, and check whether the measured ΔA_i is proportional to the predicted gradient. Only the direction matters, not the magnitude—the unknown Θ_{ii} absorbs all per-neuron scaling. A failure of proportionality would rule out not just backpropagation but *any* weight-space gradient rule in a network wide enough for diagonal dominance.

Caveat: activity descent is necessary but not sufficient for weight descent. The converse is weaker: observing that activities follow their loss gradient constrains the projection of $\Delta\mathbf{W}$ onto the row space of the Jacobian $\partial A_i/\partial W_\alpha$, but says nothing about the (typically vast) null space of weight changes that leave activities unchanged on the current input. Activity-level gradient descent is therefore consistent with many different weight-level learning rules, of which backpropagation is the minimum-norm solution. In short, weight descent implies activity descent (in wide networks), but activity descent does not uniquely identify the weight-level rule. This asymmetry is a feature, not a bug: it means the test can *falsify* gradient-based learning, even though it cannot confirm a specific synaptic rule.

Layer-local formulation of the kernel

The global kernel Θ couples every neuron to every other neuron in the network. But the feedforward structure of the network permits a more local description, in which activity dynamics are computed layer by layer. The key observation is that for a neuron in layer ℓ, the loss gradient $\partial L/\partial A_i^{(\ell)}$ is a *sufficient statistic* for all downstream activity: it compresses everything that layers $\ell{+}1,\ldots,L$ contribute to the weight updates that affect neuron i.

Separating own-layer and propagated contributions. The activity of neuron i in layer ℓ depends on two things: the weights $W^{(\ell)}$ feeding into it, and the activities $A^{(\ell-1)}$ of the previous layer. Linearising in the weight update gives

$$\Delta A_i^{(\ell)} \;=\; \underbrace{\sum_j \frac{\partial A_i^{(\ell)}}{\partial W_{ij}^{(\ell)}}\,\Delta W_{ij}^{(\ell)}}_{\text{own-layer weights}} \;+\; \underbrace{\sum_j \frac{\partial A_i^{(\ell)}}{\partial A_j^{(\ell-1)}}\,\Delta A_j^{(\ell-1)}}_{\text{propagated from layer below}}\,. \tag{6}$$

The own-layer term is diagonal. Because $W_{ij}^{(\ell)}$ feeds exclusively into neuron i, no other neuron in layer ℓ appears. Substituting $\Delta W_{ij}^{(\ell)} = -\eta\,\sigma'(z_i^{(\ell)})\,\frac{\partial L}{\partial A_i^{(\ell)}}\,A_j^{(\ell-1)}$ and $\frac{\partial A_i^{(\ell)}}{\partial W_{ij}^{(\ell)}} = \sigma'(z_i^{(\ell)})\,A_j^{(\ell-1)}$, the own-layer contribution evaluates to

$$\Delta A_i^{(\ell)}\big|_{W^{(\ell)}} \;=\; -\eta\,\big[\sigma'(z_i^{(\ell)})\big]^2\,\|\mathbf{A}^{(\ell-1)}\|^2\,\frac{\partial L}{\partial A_i^{(\ell)}}\,. \tag{7}$$

This is purely *diagonal*: each neuron's own-layer contribution depends only on its own loss gradient, with a positive scalar that depends on local quantities (the activation derivative and pre-synaptic activity norm). No same-layer kernel is needed.

The propagated term couples adjacent layers. Writing $J_\ell = \partial \mathbf{A}^{(\ell)}/\partial \mathbf{A}^{(\ell-1)}$ for the inter-layer Jacobian, the second term in Eq. **??** is $J_\ell \, \Delta \mathbf{A}^{(\ell-1)}$, which couples neuron i in layer ℓ to the activity changes of its pre-synaptic neurons through the forward weights:

$$\Delta A_i^{(\ell)}\big|_{\text{earlier}} \;=\; \sigma'(z_i^{(\ell)}) \sum_j W_{ij}^{(\ell)} \, \Delta A_j^{(\ell-1)} \,. \tag{8}$$

This term carries the effect of earlier-layer weight updates forward through the network. It is the *sole* source of off-diagonal coupling in the per-layer dynamics.

Layer-recursive activity dynamics. Combining Eqs. **??**–**??** and writing $D_\ell = \mathrm{diag}\big([\sigma'(z_i^{(\ell)})]^2 \, \|\mathbf{A}^{(\ell-1)}\|^2\big)$ gives a compact recursion:

$$\boxed{\Delta \mathbf{A}^{(\ell)} \;=\; -\eta \, D_\ell \, \nabla_{\mathbf{A}^{(\ell)}} L \;+\; J_\ell \, \Delta \mathbf{A}^{(\ell-1)} \,,} \tag{9}$$

with base case $\Delta \mathbf{A}^{(1)} = -\eta \, D_1 \, \nabla_{\mathbf{A}^{(1)}} L$ (the input does not change, so there is no propagated term). At every layer, the dynamics split into a diagonal "self" term driven by the local loss gradient, and a propagated term carrying upstream changes forward.

A per-layer kernel via the sufficient statistic. In a feedforward network, $\nabla_{\mathbf{A}^{(\ell-1)}} L = J_\ell^T \, \nabla_{\mathbf{A}^{(\ell)}} L$: the loss gradient at layer $\ell-1$ is fully determined by the loss gradient at layer ℓ and the inter-layer Jacobian. Assuming inductively that $\Delta \mathbf{A}^{(\ell-1)} = -\eta \, \Phi^{(\ell-1)} \nabla_{\mathbf{A}^{(\ell-1)}} L$ and substituting, the recursion closes into

$$\Delta \mathbf{A}^{(\ell)} = -\eta \, \Phi^{(\ell)} \, \nabla_{\mathbf{A}^{(\ell)}} L \,, \tag{10}$$

where the per-layer kernel satisfies a Lyapunov-like recursion:

$$\Phi^{(\ell)} = D_\ell + J_\ell \, \Phi^{(\ell-1)} \, J_\ell^T \,, \qquad \Phi^{(1)} = D_1 \,. \tag{11}$$

One can verify that $\Phi^{(\ell)}$ equals the within-layer block $\Theta^{(\ell,\ell)}$ of the global kernel. The decomposition is exact, not an approximation.

Structural advantages. The layer-local formulation makes three things transparent that the global Θ obscures. *First*, own-layer weights never introduce same-layer coupling—D_ℓ is always diagonal. *Second*, all off-diagonal structure in $\Phi^{(\ell)}$ originates from the propagated term $J_\ell \, \Phi^{(\ell-1)} \, J_\ell^T$, i.e. from earlier-layer weight updates echoing forward through the network. *Third*, the diagonal-dominance argument acquires a recursive flavour: at each layer, a fresh diagonal D_ℓ is added, while the propagated kernel $J_\ell \, \Phi^{(\ell-1)} \, J_\ell^T$ is decorrelated by the random projections in J_ℓ. Width helps at every stage of the recursion, not just globally.

Methods. Figures 1–2 numerically verify Eqs. 2–5 in a 3-hidden-layer ReLU MLP trained with online SGD ($\eta = 0.005$) on a 2-class image dataset (4×4 pixel images of horizontal vs. vertical bars, 20 per class, input dimension 16, output dimension 2). Dead ReLU neurons (pre-activation ≤ 0 on the sampled input) are excluded from all analyses. At regular intervals during training, we compute the full activity Jacobian $\partial A_i/\partial W_\alpha$ and the kernel Θ_{ik}, then compare the predicted activity change ΔA from the chain-rule identity (Eq. 2), the kernel equation (Eq. 3), and the diagonal approximation (Eq. 5) against the actual change measured after a single gradient step. Figure 1 shows detailed diagnostics at widths 8 and 48. Figure 2 sweeps width from 4 to 128 (3 random seeds per width, each trained for 2000 SGD steps) to quantify how the diagonal approximation improves with width.

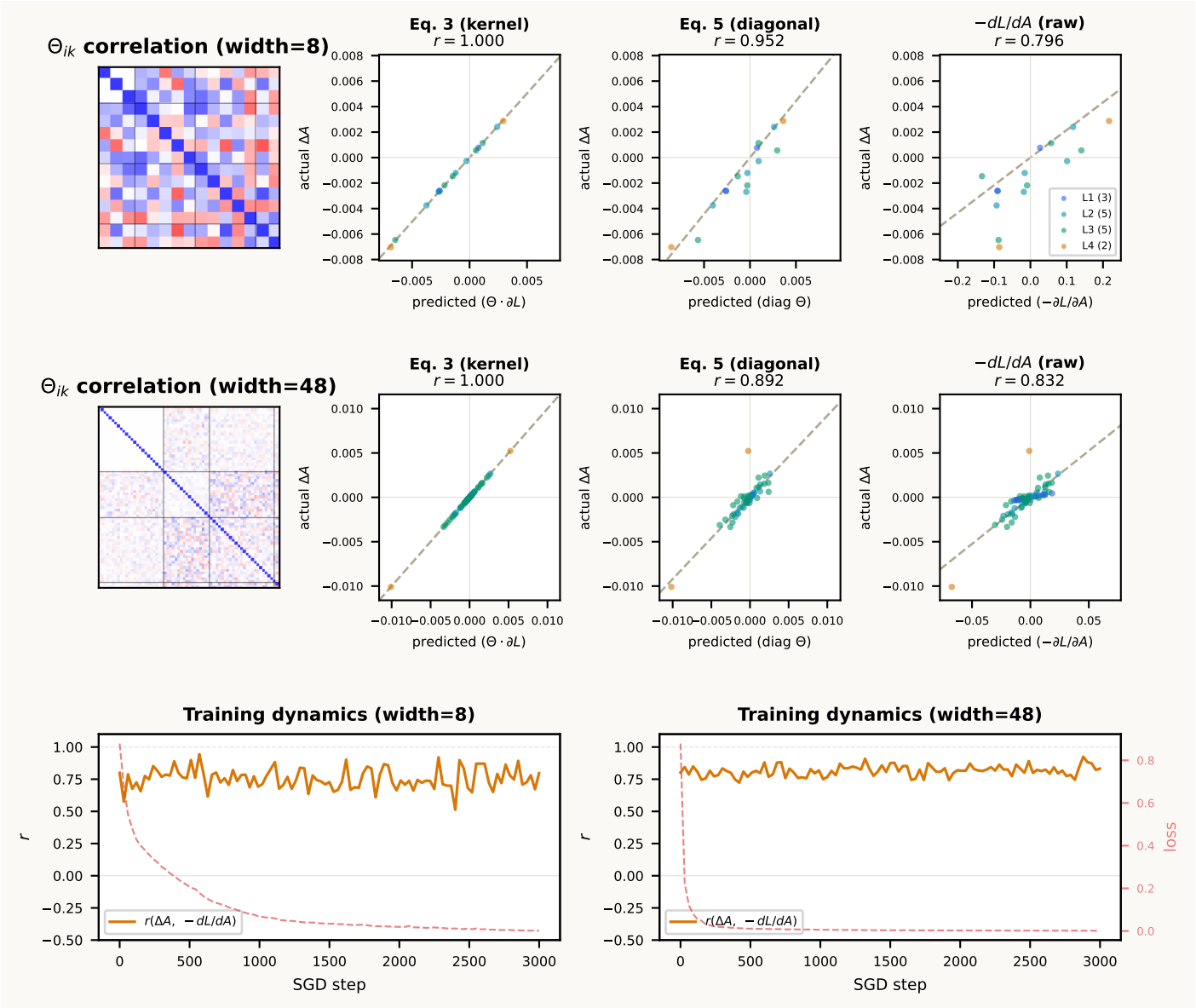

Figure 1: **In a wide network doing weight descent, activity changes closely match activity-space gradient descent.** Top row: width = 8. Middle row: width = 48. Left: normalised Θ_{ik} heatmaps (white = uncorrelated, blue = correlated). Scatter panels: predicted vs. actual ΔA for the full kernel (Eq. 3), the diagonal approximation (Eq. 5), and raw $-\partial L/\partial A$. Dead ReLU neurons excluded. Bottom row: $r(\Delta A, -\partial L/\partial A)$ and loss over training.

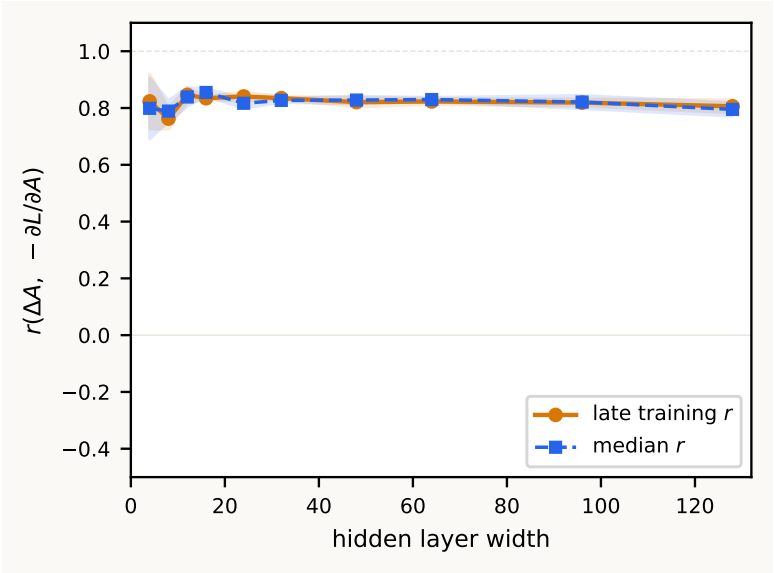

Figure 2: **The wider the network, the more faithfully activity changes track the activity-space loss gradient.** Pearson r between actual ΔA and $-\partial L/\partial A$ as a function of hidden-layer width. Dead ReLU neurons excluded. Shaded regions: ± 1 s.d. across seeds.